AI-Enhanced Segregation and Visual Quality Inspection of Earthen Ceramics Using Image Processing

MADHUMITHA V
23BMH1019

School of Mechanical Engineering
Vellore Institute of Technology
Chennai, India

MITHUNVEL K L
23BMH1029

School of Mechanical Engineering
Vellore Institute of Technology
Chennai, India

KIRAN S
23BMH1063

School of Mechanical Engineering
Vellore Institute of Technology
Chennai, India

***Abstract*—This document outlines the development and testing of a low-cost, AI-powered visual inspection system for detecting and classifying surface defects in earthen terracotta tiles. Designed to replace manual quality checks, the system utilizes lightweight deep learning models, such as MobileNetV2 and YOLOv5, running on a Raspberry Pi with a dedicated camera inside a fixed-lighting enclosure. This setup ensures consistent and repeatable image capture. The captured images undergo preprocessing steps including brightness adjustment, background normalization, and noise reduction. Classification, depending on dataset characteristics, will be performed using selected deep learning models or a K-Nearest Neighbours (KNN) classifier, categorizing flaws like cracks, chips, stains, surface roughness, and color variation. This solution addresses the inconsistencies of manual inspection, aiming to provide a portable, scalable, and cost-effective system that improves defect detection, ensures consistent quality evaluation, reduces inspection costs, and integrates AI into the production process, thereby supporting Industry 4.0 in heavy clay tile manufacturing.**

Keywords—AI, visual inspection, deep learning, defect detection, automated inspection, embedded systems, terracotta tiles, Raspberry Pi, MobileNetV2, YOLOv5

I. INTRODUCTION

n the heavy clay production line at Sree Murugan Tile Works, tile quality is the deciding factor between a finished product that meets customer expectations and one that must be scrapped. The tiles produced here are earthen terracotta products, and while they fall under the broad category of ceramics, they differ from glazed wall or floor tiles in both appearance and manufacturing. Every batch must be checked for cracks, chips, stains, uneven surfaces, and other imperfections before shipment.

For years, this inspection has been done by skilled workers who rely on their experience to spot visible flaws. They examine each tile for uniform surface texture, correct dimensions, and colour consistency. While effective in many cases, this process is not immune to human limitations. Fatigue, subjective judgement, and the repetitive nature of the work can cause inconsistencies, allowing occasional defective tiles to slip through [1], [2], [3]. In a high volume production setting, sustaining the same level of attention throughout the shift is difficult, and accuracy can drop as the day progresses.

As manufacturing shifts toward Industry 4.0, artificial intelligence (AI) and computer vision have emerged as practical tools for automating visual inspection [4], [5]. These systems can capture and analyse images at high speed, applying the same evaluation criteria for every tile. Advances in deep learning (DL), especially Convolutional Neural Networks (CNNs) and efficient detection models such as YOLO [10] and MobileNetV2 [11], now make it possible to deploy real-time inspection even on compact, low-power devices.

Previous research in the ceramics industry demonstrates that AI-based systems can reliably detect defects like cracks, chips, stains, surface roughness, and subtle colour variations under real-world factory conditions [1], [2], [4], [6]. Accuracy improves significantly when the models are trained with well prepared datasets [7], [8]. However, persistent challenges remain, some defect types occur less frequently, making it harder for the model to learn them, and varying surface textures or changes in lighting can affect detection reliability [9].

The project, “AI-Enhanced Segregation and Visual Quality Inspection of Earthen Ceramics Using Image Processing”, is being developed specifically for the 230 × 230 mm terracotta tiles produced at Sree Murugan Tile Works. The inspection system is based on a Raspberry Pi with a Raspberry Pi Camera Module, mounted inside a custom-built wooden enclosure with fixed LED lighting. This design ensures consistent image capture by blocking out changes in the ambient light.

Each tile will be placed in the enclosure, photographed, and processed through a pipeline that includes brightness correction, background normalisation, and noise reduction. The processed images will then be classified by lightweight deep learning models . Currently MobileNetV2 and YOLOv5 are being evaluated for accurate, real-time defect detection [4], [10], [11].

The aim is to produce a low-cost, portable, and scalable inspection solution that works in both large factories and smaller facilities that lack access to expensive automated quality control. By integrating AI directly into the production workflow, the system is expected to improve defect detection rates, reduce inspection costs, and maintain consistent product quality, supporting the broader use of AI-based inspection in heavy clay tile manufacturing [7], [8], [9], [17]..

II. Literature Review

1.Automated Defective Ceramic Tiles Classification using Image Processing Techniques

The manual inspection of earthen ceramic tiles is slow, inconsistent and unable to meet the production demand and quality.Traditional image processing techniques, such as morphological operations, Sobel edge detection, and thresholding, have been applied to detect defects like cracks and chipped corners. Texture based features including LBP, wavelet transforms and Gabor filters, improve classification accuracy but they require careful tuning.
Machine learning methods such as SVM have shown great results when combining the texture, shape, and color of the tile, while wavelet-based PCA and GLCM have been effective in textile defect detection, suggesting cross-domain applicability.
Deep learning approaches, including YOLOv5 achieve high accuracy but they require a huge dataset. While for the smaller datasets with 100 to 500 samples the KNN classifier has shown high accuracies(93.34% for 150 samples) in ceramic defect detection and classification.

While existing work shows that traditional image processing, texture-based features, and machine learning (like SVM) can detect ceramic defects, and deep learning methods (like YOLOv5) achieve high accuracy with large datasets, there is limited research on real-time, industrial-scale ceramic defect detection systems that work effectively with small datasets, handle diverse shapes/sizes of ceramics, and adapt to factory constraints such as lighting, dust, and subtle defects. Most prior studies either focus on post-manufacturing inspection, specific defect types, or other materials, leaving a gap for a reproducible, cost-effective, real-time solution tailored for ceramic manufacturing conditions.

2. Deep learning-based automated tile defect detection system for Portuguese cultural heritage buildings

Ganhão, Bento, and Marques created a deep learning model to find defects in the Portuguese heritage ceramic tiles. They used a dataset of 4,352 annotated images and samples that included defects like glaze detachment, cracks, lacunae and craters. The computer vision YOLOv7 was used for multi-class detection, reaching about 72% precision, 61% recall, and 64% mAP. Lacunae was identified as the most accurate while craters were found with least accuracy .The study shows the promise of AI in inspecting heritage tiles, but detecting small defects is still a challenge.

This study has several gaps that future work can fill. The system had reduced accuracy in detecting small or subtle defects like craters. This was partly because of dataset imbalance and the visual complexity of patterned tiles, which often led to misclassification. Environmental variations were only somewhat represented, which limited robustness under extreme lighting or weather conditions. Also, the approach was tested only on Portuguese azulejos, with no validation for other types of ceramics. It also lacked evaluation for real-time, on-site inspection performance

3. Surface Defect Classification in Real-Time Using Convolutional Neural Networks

The paper “Surface Defect Classification in Real-Time Using Convolutional Neural Networks” by Selim Arikan, Kiran Varanasi, and Didier Stricker addresses the challenges of traditional industrial surface inspection systems, which rely on hand-crafted features and struggle with real-time performance and limited, domain-specific datasets. The authors develop specialized CNN architectures,such as SurfNet, MultiVis, and FastInf that deliver both high accuracy and low inference time, achieving up to 99.7% accuracy on their proprietary dataset and 99.5% on the public NEU defect dataset. They compile a new dataset of over 22,000 labeled images from diverse surface materials and tackle the extreme class imbalance problem, common in defect inspection, by combining classical augmentation techniques (rotation, flipping, resizing) with GAN-based methods (pix2pix, CycleGAN). Compared to a C5.0 decision-tree classifier using more than 400 hand-crafted features, their models significantly outperform in accuracy and speed, with FastInf processing images in just 0.3 ms. Despite these impressive results, the models are tested only on their dataset and NEU, leaving questions about generalization to different surface types and real-world production-line conditions, where noise, motion blur, and varying lighting may affect performance. The approach focuses solely on classification rather than defect localization, does not deeply address deployment on low-power embedded systems, and the proprietary dataset remains unavailable, limiting reproducibility and wider adoption

4. Lightweight Tile Defect Detection Using MobileViTv3: Performance, Limitations, and Practical Considerations

The paper “Lightweight Tile Detection Algorithm Using MobileViTv3 and Multi-Scale Feature Fusion” by J. Huang, Z. Wang, and Y. Liu introduces a smart, lightweight method for detecting tile defects that balances accuracy with efficiency. Built on a MobileViTv3 backbone, the approach captures both fine local details and broader contextual information while keeping computation low. A Light-BiFPN module improves multi-scale feature fusion, a normalization-based attention module helps the model focus on actual defect areas while ignoring background distractions, and a dynamic-focused bounding box regression loss boosts localization precision. In tests, the method reached a 71.9% mAP 3.1% higher than the baseline while cutting parameters by 26.2% and computation by 20.9%, making it promising for use in resource-limited environments. Still, the research is based on a single tile defect dataset, so it’s unclear how well it would handle different materials, lighting conditions, or unusual defect patterns. The paper also doesn’t go into detail about the variety of defects covered, nor does it test real-time performance on actual edge devices. Issues like camera noise, surface reflections, or motion blur common in industrial settings aren’t explored, and there’s limited discussion about judging the severity of defects or pinpointing their exact boundaries, both of which could be crucial in real-world quality control.

5. A Review of Haralick's 1973 GLCM Texture Features: Contributions, Gaps, and Lasting Impact

The 1973 paper "Textural Features for Image Classification" by R. M. Haralick, K. Shanmugam, and I. Dinstein was a groundbreaking step in computer vision, introducing what we now call the Gray-Level Co-occurrence Matrix (GLCM) to capture patterns in how pixel values relate to each other. From these matrices, the authors extracted statistical texture features such as contrast, energy, entropy, and correlation, then used simple decision rules to classify texture-rich images like sandstone photomicrographs, aerial photos, and satellite images. The approach delivered strong results for its time, with accuracies around 89% for photomicrographs, 82% for aerial images, and 83% for satellite imagery proving that texture could be a powerful clue in image classification. Still, the work had its limitations. It focused entirely on GLCM-based statistics without exploring other texture descriptors like Local Binary Patterns or wavelets, and it relied on basic classifiers that might miss more complex patterns. The datasets, while varied, were small and specific, so it's unclear how well the method would handle larger, noisier, or more diverse data. There was also no testing for robustness against noise, lighting changes, scale variations, or distortions,common issues in real-world imagery and the potentially heavy computation needed for GLCMs wasn't addressed. Even with these gaps, the paper laid the foundation for decades of research in texture analysis and still influences image processing today.

6. A Review of Vapnik's The Nature of Statistical Learning Theory: Core Concepts, Gaps, and Lasting Impact

Vladimir Vapnik's 1995 book The Nature of Statistical Learning Theory is a landmark in the history of machine learning, laying the foundation for how we understand and approach statistical learning today. It frames learning as the challenge of estimating functions from real-world data and introduces influential ideas like empirical risk minimization (ERM), structural risk minimization (SRM), and the VC dimension, which measures how complex a model can be. Vapnik's work also explains how to strike a balance between fitting the training data and keeping the model simple enough to generalize well, forming the theoretical backbone of Support Vector Machines (SVMs). The book is written to be clear and approachable, offering deep insights without overwhelming readers with heavy proofs. That said, its focus is largely on binary classification and neat, controlled settings, leaving some unanswered questions when it comes to modern, high-capacity models like deep neural networks, which often behave in ways SRM can't fully explain. It also doesn't dive deeply into the computational hurdles of applying SRM or calculating VC dimensions, and its direct relevance to messy, noisy, or massive real-world datasets can be limited. Still, this work remains a classic—shaping decades of research and continuing to influence how we think about the balance between accuracy and simplicity in machine learning.

7. Roboflow's Practical Guide to Building a Surface Defect Detection Pipeline: Strengths and Limitations

Roboflow's 2023 blog post "Building a Surface Defect Detection Pipeline" is a hands-on, beginner-friendly guide to creating a real-time defect detection system, using ceramic tiles as the example. It walks readers through the full process, from collecting images with tools like Roboflow Collect or Roboflow Universe, to annotating defects such as edge chips, holes, and surface lines with Smart Polygon, generating dataset versions, training a model from an MS COCO checkpoint, and finally deploying it on various platforms like NVIDIA Jetson, Raspberry Pi, Luxonis OAK, web browsers, iOS, and Dockerized CPU setups. The post's biggest strength is its simplicity and step-by-step clarity, making it easy for anyone to get a working prototype up and running. However, it leaves some questions unanswered , there's little explanation of how to choose the best model architecture or tweak hyperparameters, no performance metrics like accuracy or speed, and no guidance on testing under tricky real-world conditions such as poor lighting or glare. It also doesn't go deep into how to adapt the pipeline for other materials, handle changing defect types, or maintain performance over time with continuous updates. Still, it's a great starting point for anyone who wants to quickly experiment with defect detection without getting lost in technical complexity.

8. MCAW-YOLO:An EfficientDetection Model for Ceramic Tile Surface Defects

The paper "MCAW-YOLO: An Efficient Detection Model for Ceramic Tile Surface Defects" provides a concise review of traditional and deep learning-based approaches for ceramic tile surface defect detection. The authors highlight the limitations of manual inspection including subjectivity and inefficiency, and discuss the rise of deep learning techniques, particularly the YOLO (You Only Look Once) series of object detection algorithms. YOLO-based models have shown promise by enabling rapid and accurate detection, but previous improvements often come with increased computational demands, limiting their deployment on resource-constrained devices. The paper summarizes how recent works improved feature extraction and accuracy using deeper or more complex backbone networks, multi-scale fusion, and attention mechanism.

As a response, the paper introduces the MCAW-YOLO model, which integrates a mobile-friendly vision transformer (MobileViTv3) with a lightweight feature fusion module (Light-BiFPN) and a normalization-based attention mechanism. The model also adopts a Wise-IoU loss function to improve bounding box regression, focusing on challenging samples. Through ablation and comparative experiments, the authors demonstrate that MCAW-YOLO achieves competitive accuracy and significant parameter and computation reduction compared to existing lightweight and mainstream YOLO models, making it well-suited for real-time, edge-device deployment

The paper heavily focuses on technical improvements to the YOLOv5n model but pays limited attention to the qualitative challenges of real-world defect distributions, such as unseen defect types or cross-domain generalization. The experiments are primarily conducted on a single, well-structured dataset, with only brief generalization tests on a magnetic tile dataset.

The data augmentation and preprocessing steps (such as the sliding window method) are tailored to small object detection but may introduce label noise and boundary artifacts; the impact of these choices is acknowledged but not addressed in depth.

9. QUALITY AND PROBLEMS IN THE IDENTIFICATION AND FALSIFICATION OF CERAMIC TILES

The paper titled "QUALITY AND PROBLEMS IN THE IDENTIFICATION AND FALSIFICATION OF CERAMIC TILES" appears to address expert and evaluative activities related to ceramic tile quality control and identification, likely focusing on methods to ensure tile authenticity and detect defects or falsifications. Given the source is a collection of conference materials published by Lugansk National University, it discusses contemporary issues in product expertise, including challenges with the accurate assessment of ceramic tiles in industrial and forensic contexts.
Briefly, the literature likely reviews various identification techniques and potential quality problems faced in ceramic tile manufacturing and market verification. The paper may explore how identification methods are applied to detect counterfeit tiles, assess physical properties, or control quality in production lines and trading. It would also address limitations and practical challenges in current evaluation methodologies, including gaps such as insufficient robustness against sophisticated falsifications or environmental influences.

However, limited coverage of advanced automated detection technologies like deep learning that have shown promise in related defect detection fields. A probable lack of comprehensive studies on the adaptation of identification methods under diverse industrial and supply chain conditions. Underexplored integration of identification systems with real-time quality assurance processes or digital traceability frameworks. Potential scarcity of holistic approaches combining physical, chemical, and visual assessment modalities to tackle complex falsifications effectively.

10. Automatic detection and classification of the ceramic tiles' surface defects:

The paper "Automatic detection and classification of the ceramic tiles' surface defects" presents an automatic image processing system designed for high accuracy and time-efficient detection and classification of surface defects on ceramic tiles. The study leverages advanced image processing techniques to address the challenges of identifying various defect types on tile surfaces. It focuses on improving defect recognition accuracy while maintaining computational efficiency, making it suitable for industrial applications where real-time processing is critical. The approach integrates effective feature extraction and classification methods to distinguish between different defect categories, enhancing quality control in ceramic tile manufacturing.
Gaps in paper:
Despite its contributions, the paper has some limitations. It primarily addresses defect detection under controlled imaging conditions, which might not fully capture the variability of real-world industrial environments such as varying lighting or tile surface textures. The study also largely relies on traditional image processing and classification techniques, leaving room for incorporating more recent deep learning approaches that have shown superior performance in defect detection tasks. Furthermore, the robustness of the system against complex defect patterns and its scalability to large datasets or diverse tile types are areas that could benefit from further exploration. These gaps point to opportunities for integrating more adaptive, data-driven models and evaluating system performance in more varied, real-world manufacturing settings.

11. An Enhanced Automatic Surface and Structural Flaw Inspection and Categorization using Image Processing Both for Flat and Textured Ceramic Tiles

The paper titled "Automatic Defect Detection and Classification of Ceramic Tiles" presents an image analysis and computer vision system designed to efficiently detect and classify surface defects in ceramic tiles. The research addresses the challenge of maintaining high-quality production rates in the ceramic tile industry, where manual inspection is labor-intensive, subjective, and time-consuming. The proposed method involves three key steps: image preprocessing (including acquisition, enhancement, noise reduction, and edge detection), defect detection in test images compared against reference images, and classification of six common defect types—crack, blob, pinhole, spot, corner, and edge defects. The system enhances gray-level images to improve recognition accuracy using contrast stretching, median filtering for noise reduction, and Sobel edge detection for highlighting edges. The detection decision is made by comparing the number of black pixels (representing defects) between processed test and reference images. Through this approach, the system aims to provide a rapid, consistent, and objective alternative to manual inspection, applicable during production and packaging to prevent defective tiles from reaching end-users.

Despite its contributions, the paper notes limitations and areas for future work. Notably, it does not address the detection of glaze and scratch defects, which remain unclassified with the current method. The computational efficiency has not been measured, suggesting a need for investigation into reducing processing time for practical industrial deployment. Additionally, the system relies on comparison with reference images, which may limit adaptability to varied tile designs and defect manifestations. The classification mechanism has not been fully automated in some aspects, as human vision still plays a role in defect categorization in related prior works referenced by the authors. As such, potential gaps include the need for more comprehensive defect type coverage, computational optimization, robustness to diverse tile textures and defect variations, and enhanced automation in defect classification to further improve reliability and industrial applicability. This research builds a basis for automatic ceramic tile defect detection but invites extension towards more adaptive, real-time, and inclusive quality control systems.

12. Quality Control System for Ceramic Tiles using Segmentation-based Fractal Texture Analysis and SVM

The paper "Quality Control System for Ceramic Tiles using Segmentation-based Fractal Texture Analysis and SVM" addresses the limitations of manual inspection in ceramic tile manufacturing—an industry with highly automated production lines but still heavily reliant on human judgment for quality control. Traditional visual inspection is affected by fatigue, inconsistency, and subjectivity, creating the need for automated solutions. The literature review in the paper covers various prior approaches: from Discrete Wavelet Transform with neural nets, morphological operations and probabilistic neural networks, to Local Binary Pattern (LBP) and k-nearest neighbor classifiers. These previous systems sought to optimize defect detection accuracy and reduce computation time but did not always generalize well across tile types or production environments.

The authors highlight a key methodological gap: while Segmentation-Based Fractal Texture Analysis (SFTA) has been employed in other fields (such as railway track inspection and animal identification), it has not previously been applied to ceramic tile defect detection. To address this, the study proposes a complete pipeline, including controlled image acquisition, preprocessing to isolate the tile, SFTA for feature extraction, and Support Vector Machine (SVM) for classification. The goal is to deliver high-speed, accurate quality control suitable for real-world production, with a strong emphasis on implementing a low-cost prototype and validating its performance with a sizable image dataset.

The system was tested only under a controlled environment, which may not fully represent the variability in real industrial settings. The robustness of the proposed method in less controlled, real-world scenarios remains unexamined.
The system classifies tiles as simply "good" or "bad", without attempting to specify defect types or severities, which limits its utility for process improvement or targeted quality control.

13. Assessment of Quality Control of Tiles Production in West Africa Ceramics Company, Ajaokuta, Kogi State

The paper "Assessment of Quality Control of Tiles Production in West Africa Ceramics Company, Ajaokuta, Kogi State" provides a comprehensive evaluation of quality control measures implemented in a leading Nigerian tile manufacturing plant. The study frames Total Quality Management (TQM) as essential for industrial competitiveness, aligning with ISO-9001:2015 standards to ensure consistent production quality. Quality control in the plant is managed using tools like P-charts (for monitoring defect ratios), operation characteristic curves, and mechanical resistance tests—key statistical and physical methods for assessing tile performance and batch uniformity. The research documents a significant improvement in compliance, showing a 71% implementation level for TQM in 2022; this is tracked both through numeric compliance ratings and longitudinal data since the adoption of ISO-9001:2015 certification. The study identifies specific phases in the production process (mixing, forming, glazing/firing, sorting/packaging) where quality checks are critical, employing instruments such as universal strength and hardness testers to quantify product reliability.

The authors also analyze operational data across six categories of tiles (various sizes), detailing defective rates, standards for maximum allowable defects per batch, and consumer acceptance levels most notably, the expectation of near-zero defects for market-ready tiles. The use of compliance regression and the PDCA (Plan-Do-Check-Act) cycle demonstrates the company's commitment to continuous improvement.

The study mainly evaluates the implementation of traditional TQM and ISO standards without investigating advanced automated or AI-based inspection systems, which are increasingly common in global tile manufacturing for defect prediction and real-time quality monitoring.
While the paper uses quantitative control charts and mechanical tests, it offers limited insight into specific types of defects (other than general defect counts), lacking a deeper analysis on root cause identification or tracking diverse visual/structural flaw categories.The aim of this work is to design, develop, and test an affordable AI-powered visual inspection system that can detect and classify surface defects in earthen terracotta tiles in real time. The system will replace traditional manual quality checks by using lightweight deep learning models such as MobileNetV2 and YOLOv5, running on a Raspberry Pi with a dedicated camera inside a fixed lighting enclosure to maintain consistent and repeatable image capture.
Each tile will be placed inside the enclosure and photographed under controlled LED lighting. The captured images will then be processed to adjust brightness, normalise the background, and reduce noise. Depending on the size and characteristics of the dataset, classification will be performed using either the selected deep learning models or a K-Nearest Neighbours (KNN) classifier. The classification step will not only detect flaws but also group them into defined categories, focusing on defects such as cracks, chips, stains, surface roughness, and colour variation.
This approach is intended to address the common problems with manual inspection, including fatigue, subjective judgement, and inconsistent results. The goal is to deliver a portable, scalable, and low-cost solution that can be adopted in both large manufacturing plants and small production setups. By improving defect detection rates, ensuring consistent quality evaluation, reducing inspection costs, and integrating AI into the production process, the system supports the transition toward Industry 4.0 in heavy clay tile manufacturing.

III. Objectives

The aim of this work is to design, develop, and test an affordable AI-powered visual inspection system that can detect and classify surface defects in earthen terracotta tiles in real time. The system will replace traditional manual quality checks by using lightweight deep learning models such as MobileNetV2 and YOLOv5, running on a Raspberry Pi with a dedicated camera inside a fixed lighting enclosure to maintain consistent and repeatable image capture.

Each tile will be placed inside the enclosure and photographed under controlled LED lighting. The captured images will then be processed to adjust brightness, normalise the background, and reduce noise. Depending on the size and characteristics of the dataset, classification will be performed using either the selected deep learning models or a K-Nearest Neighbours (KNN) classifier. The classification step will not only detect

flaws but also group them into defined categories, focusing on defects such as cracks, chips, stains, surface roughness, and colour variation.

This approach is intended to address the common problems with manual inspection, including fatigue, subjective judgement, and inconsistent results. The goal is to deliver a portable, scalable, and low-cost solution that can be adopted in both large manufacturing plants and small production setups. By improving defect detection rates, ensuring consistent quality evaluation, reducing inspection costs, and integrating AI into the production process, the system supports the transition toward Industry 4.0 in heavy clay tile manufacturing.

IV. Budget Table (Draft)

TABLE I. ESTIMATED BUDGET

S.No	Item	Description	Price (₹)
1	Raspberry Pi 4 Model B (4GB)	Main processing unit for AI inference	6500
2	Raspberry Pi Camera Module 3	12MP auto-focus camera	3000
3	32GB MicroSD Card (Class 10)	OS and data storage	500
4	LED Light	Consistent lighting setup	800
5	Wooden Enclosure	Camera + lighting control box	1500

V. Peer Rating

23BMH1063: KIRAN S	10
23BMH1029: MITHUNVEL K L	10
23BMH1019: MADHUMITHA V	10

Acknowledgment

We would like to express our sincere gratitude to Dr. Abhishek Rudra Pal, Professor (Computer Science) as well as the Dean of SMEC (School of Mechanical Engineering), Vellore Institute of Technology, for their invaluable guidance and support throughout this project.

We would also like to thank the co-authors for their dedicated efforts and contributions to this work.

References

[1] J. Marín, A. Molina, M. Martínez, P. M. Ortigosa, and M. P. Lucena, “A Real-Time Automated Defect Detection System for Ceramic Pieces in Industry,” Sensors, vol. 24, no. 1, p. 232, Jan. 2024. [Online]. Available: https://www.mdpi.com/1424-8220/24/1/232

[2] J. Ganhão, C. Bento, and F. Marques, “Deep Learning-Based Automated Tile Defect Detection System for Portuguese Cultural Heritage Buildings,” ResearchGate, 2024. [Online]. Available: https://www.researchgate.net/publication/381065626_Deep_learning-based_automated_tile_defect_detection_system_for_Portuguese_cultural_heritage_buildings

[3] L. Nunes, F. Soares, and H. Puga, “A Review of Automated Visual Inspection Systems for Surface Defects in Ceramics,” Computers in Industry, vol. 108, pp. 88–100, Mar. 2019.

[4] A. Prasad, M. Mukherjee, and A. Goswami, “Automated Vision Inspection in Manufacturing: A Review,” Engineering Applications of Artificial Intelligence, vol. 87, p. 103249, Jan. 2020.

[5] X. Song, H. Zhang, and J. Li, “Recent Advances in Surface Defect Detection for Industrial Products Using Computer Vision,” IEEE Access, vol. 10, pp. 49231–49252, Apr. 2022.

[6] Y. Huang, X. Li, H. Chen, and J. Zhang, “Ceramic Tile Surface Defect Detection Based on Deep Learning,” Electronics, vol. 14, no. 12, p. 2365, Jun. 2023.

[7] M. Shaikh, S. Shaikh, and M. Chavan, “Review on Deep Learning Approaches to Detect Defects on Ceramic Tiles,” Materials Today: Proceedings, 2023.

[8] Y. Zhang, Z. Sun, and H. Wu, “A Survey of Surface Defect Detection with Computer Vision,” Computers in Industry, vol. 129, p. 103438, 2021.

[9] S. B. Othman, M. B. Ayed, and M. Atri, “Vision-Based Defect Detection in Industrial Settings: A Deep Learning Perspective,” Sensors, vol. 22, no. 14, p. 5407, Jul. 2022.

[10] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “YOLOv4: Optimal Speed and Accuracy of Object Detection,” arXiv preprint arXiv:2004.10904, Apr. 2020.

[11] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2018, pp. 4510–4520.

[12] O. Russakovsky et al., “ImageNet Large Scale Visual Recognition Challenge,” International Journal of Computer Vision, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[13] OpenCeramics, “Open Datasets for Ceramic Surface Defect Detection,” 2022. [Online]. Available: https://openceramics.org/dataset

[14] Roboflow, “Building a Surface Defect Detection Pipeline,” Roboflow Blog, 2023. [Online]. Available: https://blog.roboflow.com/surface-defect-detection/

[15] V. Vapnik, The Nature of Statistical Learning Theory. Springer, 1995.

[16] R. M. Haralick, K. Shanmugam, and I. Dinstein, “Textural Features for Image Classification,” IEEE Transactions on Systems, Man, and Cybernetics, vol. SMC-3, no. 6, pp. 610–621, Nov. 1973.

[17] J. Huang, Z. Wang, and Y. Liu, “Lightweight Tile Detection Algorithm Using MobileViTv3 and Multi-Scale Feature Fusion,” Applied Sciences, vol. 13, no. 21, p. 12057, Oct. 2023.